

Sabrina Shirazi

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CURRICULUM VITAE (Updated 1/2021)

EDUCATION

2016-Anticipated June 2021

University of California, Santa Cruz.

Ph.D. – Ecology and Evolutionary Biology

Advisor- Dr. Beth Shapiro

Committee- Dr. Rachel Meyer, Dr. Duane Froese, Dr. Ingrid Parker

Fellowships- GAANN Fellow, PBK Scholar

Certificate- Inclusive Inquiry STEM Education

2016-2019

University of California, Santa Cruz.

M.A. - Ecology and Evolutionary Biology

Advisor- Dr. Beth Shapiro

Thesis: Developing environmental DNA methods to reconstruct modern and ancient ecosystems

2012-2016

University of Maryland, College Park.

B.S.- Individual Studies Program

Combines: Genetics, Evolution, Anthropology, and Bioinformatics

Thesis: Island Gigantism in *Peromyscus* on the California Channel Islands

B.A.- Anthropology (Environmental/Ecological focus). **with Honors

Thesis: Using Environmental DNA from Archaeological Sites to Recreate the Paleoecosystem of Santa Cruz Island

Honors- **Phi Beta Kappa** (Gamma Chapter), and Deans list in all nine semesters.

GPA 3.8/4 in 153 credits. Minor- Archaeology

GRANTS/SCHOLARSHIPS

2020	Phi Beta Kappa Northern CA Regional Graduate Scholarship- Norall award	(\$7,500)
2020	PEO Chapter LP, Saratoga Scholarship	(\$500)
2019	UCSC EEB Travel Support	(\$1,200)
2019	Norris Center Science and Art Residency	(\$1,500)
2018	UC Conservation Genomics Consortium for Conservation Genomics Workshop	(\$225)
2016	GAANN support for Professional Development Program	(\$4,070)
2016-2019	GAANN fellowship (3 years of stipend and tuition)	
2016	U. of Maryland, Individual Studies Program – Travel award	(\$700)
2016	Smithsonian Minority Awards Program - Borneo Bird Genetics	(\$4,000)
2015	U. of Maryland, Individual Studies Program- summer research award	(\$500)
2015-2016	Smithsonian Minority Awards Program- Gray Fox Genetics and Nanopore	(\$19,500)
2015	Smithsonian Castle- small research support grant	(\$500)
2015	Universität Leipzig - PEB Workshop travel award	(€925EUR)
2015	U. of Maryland, Individual Studies Program–workshop attendance award	(\$1,000)
2015	U. of Maryland, Anthropology Department- workshop attendance award	(\$500)
2015	U. of Maryland, BSOS – workshop attendance award	(\$500)
2014	U. of Maryland, Beyond the Classroom Program. Transportation funding	(\$250)

2014	U. of Maryland, BSOS/ANTH Department – workshop attendance award	(\$500)
	<i>Funding obtained for the Society for Conservation Biology: Santa Cruz Chapter, as Vice President</i>	
2017	UCSC ODEI- Diversity and Inclusion Grant	(\$1000)
2017	Society for Conservation Biology Chapter Grant for conference audio support	(\$750)
2017	UCSC Graduate Student Administration- funds for conference room rental	(\$680)

PUBLICATIONS

- Shirazi, S.** Meyer, R., Green, R., Shapiro, B.. (2021). Exploring PCR Replication in Environmental DNA Metabarcoding. In Prep.
- Shirazi, S.**, Broomand, N., Metcalfe, M.,...Ives, J.. (2021). Genomic-based sex identification of moccasins and hide from Promontory Cave, UT. In Prep.
- Reding, D., Caseñada-Rico, S., **Shirazi, S.**, Maldonado, J.. (2021). Mitochondrial genomes of the USA distribution of gray fox (*Urocyon cinereoargenteus*) reveals a major phylogeographic break at the Great Plains Suture Zone. *Frontiers in Ecology and Evolution*. In Review.
- Cronin, C., ..., **Shirazi, S.**, ... Zavaleta, E.. (2021). Implementing Anti-Racist Pedagogy and Practice in Ecology, Evolution, & Conservation Biology Departments. *Nature Ecology and Evolution*. In Review.
- Curtin, L, D’Andrea, B., **Shirazi, S.**,...Bradley, R.. (2021). Sedimentary DNA and molecular evidence for pre-Norse occupation of the Faroe Islands. *Science Advances*. In Review.
- Wygall, B., ...**Shirazi, S.**, ... Tullo, D.. (2021). A Late Pleistocene Hair Raises Questions about Human-Mammoth Interactions in Eastern Beringia. *Environmnetal Archaeology*. In Review
- Castro, L.,..., **Shirazi, S.**, ... Quiroga, S.. (2021). Metabarcoding Meiofauna Biodiversity Assessment in Four Beaches of Northern Colombia: Effects of Sampling Protocols and Primer Choice. *Hydrobiologia*. In Review.
- Lin, M., ..., **Shirazi, S.**, ...Meyer, R.. (2021). A Biodiversity Map of California Derived from Environmental DNA Metabarcoding and Earth Observation. *Ecological Applications*. In Review.
- Metcalfe, J., Ives, J., **Shirazi, S.**...Shapiro, B.. (2021). Isotopic evidence for long-distance connections of the AD thirteenth century Promontory Caves occupants. *American Antiquity*. In Press.
- Meyer, R., Curd, E., Schweizer, T., Gold, Z., Ramos, D., **Shirazi, S.**....(2019). The California Environmental DNA “CALeDNA” Program. [California Agriculture](#).
- Castañeda, S., Tsuchiya, M., **Shirazi, S.**, Hofman, C., Redding, D., Rick, T., Edwards, C., and Maldonado, J. *Insights into the evolutionary history of gray foxes revealed by mitogenomes from museum specimens*. In Prep.
- Shirazi, S.**, Rick, T., Erlandson, J., Hofman, C.. (Dec. 2017) *A Tale of Two Mice: A Trans-Holocene Record of Peromyscus nesodytes and P. maniculatus at Daisy Cave, San Miguel Island, California*. [The Holocene](#).
- West, C., Hofman, C., Ebbert, S., Martin, J., **Shirazi, S.**, Dunning, S., Maldonado, J.. (Nov. 2016) *Integrating archaeology and ancient DNA to address invasive species colonization in the Gulf of Alaska*. [Conservation Biology](#).

CONFERENCE PRESENTATIONS

- Curtin, L, D'Andrea, B., **Shirazi, S.**,... Bradley, R..(2020).Sedimentary DNA and molecular evidence for pre-Norse occupation of the Faroe Islands. American Geophysical Union Conference. Virtual.
- Shirazi, S.**, Cassatt-Johnstone, M., Vershinina, A., Oppenheimer, J., Froese, D., Shapiro, B.. (December 2019). "Long-Term Ecological Change in the Klondike Exposed with Permafrost eDNA". 100th American Geophysical Union Conference. San Francisco, CA.
- Wygall, B., Krasinski, K., Holmes, C., Crass, B., Smith, K., McMahan, D., Metcalfe, J., **Shirazi, S.**... (April 2020). "The Holzman Site: 14,000-years of Human Activity along Shaw Creek, Alaska". Society for American Archaeology 85th Annual Meeting. Austin, TX.
- Lin, M., **Shirazi, S.**, ... Wayne, R.. (June 2019). "Comparison of capture array, metabarcoding, and shotgun sequencing in recovering mammalian eDNA from contemporary soil samples". International Barcode of Life conference. Trondheim, Norway.
- Lin, M., ... **Shirazi, S.**, ... Wayne, R.. (June 2019). "Mapping biodiversity across California using remote sensing and community science eDNA survey". International Barcode of Life conference. Trondheim, Norway.
- Shirazi, S.**, Meyer, R., Shapiro, B.. (Nov. 2018). "eDNA: PCR Replicate Test". UC Conservation Genomics Consortium.
- Shirazi, S.** (Dec. 2017). "CaleDNA from Mountain Areas". UC Conservation Genomics Consortium eDNA Workshop.
- Shirazi, S.**, Supple, M., Garza, J., Pearse, D., Shapiro, B.. (Aug. 2017). "Conservation Genomics of California Trout". UC Conservation Genomics Consortium.
- Shirazi, S.**, Rick, T., Erlandson, J., Hofman, C.. (May 2017). "A Tale of Two Mice: A Trans-Holocene Record of *Peromyscus nesodytes* and *P. maniculatus* at Daisy Cave, San Miguel Island, California.". UCSC 13th annual Graduate Research Symposium.
- Shirazi, S.**, Rick, T., Erlandson, J., Hofman, C.. (April 2017). "A Tale of Two Mice: A Trans-Holocene Record of *Peromyscus nesodytes* and *P. maniculatus* at Daisy Cave, San Miguel Island, California.". UCSC EEB Grad-Student Symposium.
- Hofman, C., Rick, T., **Shirazi, S.**, Maldonado, J., (Sept. 2016). "Ancient Invasive Species? The Origins of the Endemic Mammals of the California Channel Islands" at the International Symposium for Biomolecular Archaeology. Oxford, England.
- Hofman, C., Maldonado, J., **Shirazi, S.**, Rick, T.. (June 2016). "*Tracking Translocations: Interdisciplinary approaches to investigating the origins of the California Channel Island Mammals*" at the American Society of Mammalogists Annual Conference. Minneapolis, MN.
- Castaneda-Rico, S., Tsuchiya, M., **Shirazi, S.**, Hofman, C., Reding, D., Rick, T., Maldonado, J., (June 2016). "*Insights into the evolutionary history of Mesoamerican Gray Foxes revealed by mitogenomes from museum specimens*" at the American Society of Mammalogists Annual Conference. Minneapolis, MN.
- Shirazi, S.**, Hofman, C., Rick, T., Maldonado, J., (April 2016). "*Santa Cruz Island Paleoecosystem Reconstruction Using Environmental DNA*" at the Society for American Archaeology Annual Conference. Orlando, FL.
- Hofman, C., Rick, T., **Shirazi, S.**, Maldonado, J., (April 2016). "*Archaeogenomics and the mammals of CA Channel Islands*" at the Society for American Archaeology Annual Conference. Orlando, FL.

- Shirazi, S., Hofman, C., Rick, T., Maldonado, J., (2016).** “*Using Environmental DNA at Archaeological Sites to Recreate Paleoecosystems*” at the University of Maryland Anthro+ Conference. College Park, MD.
**** Conference Paper Prize Winner.**
- Shirazi, S. (2015).** “*Capturing of Mitogenomes for Phylogeographic Analysis*” at Smithsonian Conservation Biology Institute 4th Annual Student Research Symposium. National Zoological Park.
- Shirazi, S. and Damann, F. (2015).** “*Osteological Healing and Medical Practice During the Civil War*” at the University of Maryland Anthro+ Conference. College Park, MD.
- Shirazi, S. (2014).** “*Femoral healing during the Civil War*” at the University of Maryland Anthropology Student Association Undergraduate Research Symposium. College Park, MD.
- Shirazi, S. and Hofman, C. (2014).** “*Nanopore DNA sequencing at the Center for Conservation and Evolutionary Genetics, SCBP*” at Smithsonian Conservation Biology Institute 3rd Annual Student Research Symposium. Front Royal, VA.

TEACHING EXPERIENCE AND LEADERSHIP

- Nov. 2018 - Present **UCSC CALeDNA Point Scientist**
 Developed and lead workshops to teach basic scientific theory skills and molecular lab work techniques to diverse groups of UCSC and UCLA students.
 Mentored undergraduate researchers. Hosted community outreach events on natural reserves.
- Spring 2020 **Course TA:** Biodiversity in the Age of Humans
- July 2019 **Course Instructor:** CaleDNA summer research institute. UCSC
- July 2018 **Course Instructor:** CaleDNA summer research institute. UCSC
- June 2018 **Course Instructor:** CaleDNA summer research institute. UCLA
- 2017 **Guest Speaker:** ‘Beringia- the Gold Mine for Ancient DNA’ Lecture for Yukon Beringia Interpretive Centre
- 2017 **Professional Development Program Participant-** 6 month program in developing inquiry, equity and inclusion based teaching skills in STEM. Developed 6 hour Ecology program for undergraduate transfer students.
- Winter 2017 **Designed Class-** designed UCSC Climate Change Seminar
- Fall 2016-Present **Vice President-** Santa Cruz Chapter of the Society for Conservation Biology
 -Organized [conference](#), raised funds, and promote local knowledge and action in conservation
- 2014-2015 **TA:** Introduction to Environmental/Ecological Anthropology
 University of Maryland, College Park. 3 semesters as Lab TA.
- Jan. 2016 **Guest Speaker:** Elysian Charter School, Hoboken, NJ.
 -Conservation Genetics
- Jan. 2014 **Guest Speaker:** Darfur Friends Association, Tel Aviv, Israel
 -Agricultural Influences

RESEARCH EXPERIENCE

Sep. 2016- Present

University of California, Santa Cruz

Ph.D. student

Supervisors- Dr. Beth Shapiro and Dr. Rachel Meyer

Dissertation Chapters:

- 1- Exploring PCR Replication in Environmental DNA Metabarcoding
- 2- Tracking human presence on the Faroe Islands with sedaDNA
- 3- Floral and mammalian community changes in the Klondike observed with ancient eDNA from Permafrost.
- 4- Genomic-based sex identification of moccasins and hide from Promontory Cave, UT

Various other projects including:

-Meiofaunal diversity in Colombia tracked with ARMS and eDNA, eDNA of archaeological sites in Alaska, CALeDNA projects and collaborations

Mar. 2014- July 2016

Smithsonian Center for Conservation Genomics

Research Intern

Supervisors: Dr. Courtney Hofman and Dr. Jesus Maldonado

Research Projects:

- 1-Protocol testing and Development with MinION sequencer
- 2-Analysis of human impact on the California Channel Island Fox through the utilization of both ancient archaeological and modern tissue samples
- 3-Phylogeography of the Gray Fox
- 4- California Channel Islands Shell Middens eDNA
- 5-*Integrating archaeology and ancient DNA to address invasive species colonization in the Gulf of Alaska*

Performed research across several genomics projects using Sanger, Illumina, and Nanopore sequencing platforms with modern and ancient DNA. Was involved in sequence generation and analysis from DNA extraction through library prep, sequencing, QC, statistical analysis, data curation, and publication. Engaged with the public through various outreach programs offered through the lab and Zoo. Trained visiting scientists, PhD students, and post-docs in laboratory skills.

Mar. 2014- July 2016

Smithsonian Museum of Natural History, Anthropology Department.

Research Intern

Supervisors: Dr. Courtney Hofman and Dr. Torben Rick

Research Projects:

- 1- Island Gigantism among *Peromyscus* on the California Channel Islands
- 2-Analysis of human impact on the California Channel Island Fox through the utilization of both ancient archaeological and modern tissue samples

Used genomic and morphometric tools to examine the natural history of island foxes and deer mice on the California Channel Islands.

Jan. 2014- Aug. 2015

National Museum of Health and Medicine. Silver Spring, MD.

Research Intern

Supervisor: Dr. Franklin Damann

Compiled an inventory of the museums total skeletal collection, confirming bone identity and taphonomy known and generated a database of all specimens.

Contributed to the collections imaging database of bone histology. Worked with the museum in creating and distributing various outreach activities. Assisted in the compilation of forensic profiles for case submissions. Performed a research project on the healing observed in amputated femora from soldiers injured during the Civil War

ARCHAEOLOGY EXPERIENCE AND OUTREACH

2015-2016

Archaeological Field Technician. Belle Isle, VA.

Smithsonian National Museum of Natural History.

Dr. Leslie Reeder-Myers

-Survey and excavate archaeological shell middens

2014-2015

Maryland Day Archaeology Educational Booth. University of Maryland.

-Oversee public laboratory programs

2014

Smithsonian Environmental Research Center

-Trained in magnetometer use.

-Placed archaeological grids

2013- 2015

Archaeological Lab Technician. Archaeology in Annapolis Lab.

University of Maryland. Dr. Mark Leone.

-Archive artifacts

-Catalogue artifacts

2013-2014

Archaeological Field Technician. Easton, MD.

Archaeology in Annapolis. Dr. Mark Leone.

-Phase 1,2 & 3 excavations

-Guided tours on site history and archaeological methodology

-[Received press interviews... more](#)

2013

Historical Archaeology Field School. Student.

University of Maryland, College Park. Archaeology in Annapolis.

Dr. Mark Leone.

-Certified as an archaeological field and laboratory technician

MISC. OUTREACH

Dec. 2016-Present

Phi Beta Kappa- Outreach Committee

University of California, Santa Cruz

Sep. 2016- Present

[Undergraduate-Graduate Mentoring Program](#)

University of Maryland, Department of Anthropology

-Mentor undergraduates interested in combining biology and anthropology

-Motivate students to go beyond the classroom in seeking skills and support

- 2014-2016 **Smithsonian Center for Conservation Genomics Outreach**
 -Mentor undergraduate students from UCSB underrepresented groups
 -Discuss specific applications of genomics to conservation
- 2014-2016 **Autumn Conservation Festival.** Smithsonian Center for Conservation Biology.
 -Inform children and adults on conservation genetics
 -Administer children's learning activities
- 2014- Present **Anthropology Student Association**
 2015-2016-President
 2014-2015-Vice President
 -Lead weekly meetings and events
 -Plan educational and social events
 -Commitment to student outreach and education
- 2014- 2016 [Beyond the Classroom Living and Learning Program.](#)
 -Leadership and community service with non-profits and NGO's
 -Management at Non-Profit events
 -Involvement in Civic engagement and learning activities